

Student Name: Jinpeng Zhu
Student ID: 1907843
Date: 2026-03-26



Underground Mining and Bulk Materials Handling
(MIN E – 630)
Assignment #5

1 Question 1

1.1 what is the swell?

Swell means the volume expands to larger than its original state. In mining background, it usually means the bank state mineral or rock becoming loose state due to mining activities.

$$\text{Swell} = \text{bulking factor} - 1 = 0.6$$

Thus the swell is 60%.

1.2 what is the swell factor?

The swell factor describes the increasing volume relative to bank volume. The definition equation is shown below:

$$\text{Swell factor} = \frac{1}{1 + \text{Swell}} = \frac{1}{1 + 0.6} = 0.625$$

1.3 What is the BCM being moved by the truck on each trip?

$$\text{BCM} = 40 \times 0.625 = 25$$

1.4 What is the loose density of the materials?

$$\text{Loose density} = 2800 \times 0.625 = 1750 \text{ kg/LCM}$$

1.5 During hauling, the average specific gravity of the solid is 2.7, what is the porosity of the loose materials?

Assume that the bulk materials are dry and ignore air weight.

$$1.75 = 2.7 - c(2.7 - 0)$$

$$\Rightarrow c = 0.3518$$

That is, the loose materials include 35.18% air, so the porosity is 35.18%.

1.6 What is the payload in metric tons (t)?

$$\text{Payload} = \text{capacity} \times \text{density (LCM)}$$

$$= 40 \times 1750 = 70000 \text{ kg} = 70 \text{ t}$$

2 Question 2

$$P_{\text{pump}} = 700 \text{ kW} \quad P_e = 0.75 P_{\text{pump}} = 525 \text{ kW}$$



$$d = 0.4 \text{ m}$$

$$\text{slope angle} = 10^\circ$$

$$f_l = 0.03$$

$$f_s = 0.8$$

$$Q_m = 1000 \text{ tph} = 277.78 \text{ kg/s}$$

$$\rho_s = 2100 \text{ kg/m}^3$$

$$V = 1.5 \text{ m/s}$$

$$\rho_w = 1000 \text{ kg/m}^3$$

$$P_f = \Delta P \times 20\%$$

$$a) P_e = 525 \text{ kW}$$

$$b) \Delta P = P_f + P_s + P_p + P_k \quad (\text{in hydraulic transport, } P_k \text{ is usually negligible})$$

$$P_f = \frac{f_l L \rho V^3}{2 R_h} \quad \left. \begin{array}{l} \Delta P \times 20\% \\ f_l = 0.03 \\ L = ? \\ \rho = 1000 \text{ kg/m}^3 \text{ (water)} \\ V = 1.5 \text{ m/s} \end{array} \right\} R_h = \frac{A}{\rho \nu}$$

$$\text{Because } Q_m = A v = c s A v = c s Q$$

$$\text{Volumetric } Q_s = \frac{I}{\rho} = 0.132 \text{ m}^3/\text{s}$$

$$\text{total } Q_t = A v = \frac{1}{4} \pi d^2 \times 1.5 = 0.1885 \text{ m}^3/\text{s}$$

$$\text{Water } Q_w = Q_t - Q_s = 0.0565$$

$$c = \frac{Q_s}{Q_t} \times 100 = 70\%$$

$$P_{\text{pump}} = \frac{W}{t} = \frac{\Delta P V}{t} = \Delta P Q_t = 525000 \text{ W}$$

$$\Rightarrow \Delta P = \frac{525000}{Q_t} = \frac{525000}{0.1885} = 2.78 \text{ MPa}$$

$$c) \text{ see } , \quad c = 70\%$$

d) Based on ΔP , $\Rightarrow P_f = \Delta P \times 0.2 = 557029 \text{ Pa}$

Substitute known data to P_f .

$$P_f = \frac{f L v^2}{2 R_h} = \frac{0.03 L \times 1000 \times 1.5^2}{2 \times 0.2} = 557029$$

$$\Rightarrow L = 3300.9 \text{ m}$$

Assume the flow fills the pipeline $\left\{ R_h = \frac{\frac{1}{4} \pi d^2}{\pi d} = 0.1 \right.$

$$P_f = 557029 = \frac{0.03 L \times 1000 \times 1.5^2}{0.1 \times 2} = 1650.45 \text{ m}$$

$$\Rightarrow L = 1650.45 \text{ m}$$

e) $P_s = \mu c L_h g (s-r)$

or $P_s = \Delta P - P_f - P_p$

$$P_p = g h [r + c (s-r)]$$

$$= 9.81 \times 1650.45 \times \sin(10^\circ) \times [1000 + 70.18\% (2100 - 1000)]$$

$$= 4981962 \text{ Pa}$$

$\therefore Z_0 - Z_i < 0$, P_p is ~~negative~~ negative.

$$\therefore P_s = 2785000 - 557029 + 498192$$

$$= 7209933 \text{ Pa}$$

$$b f) k = \frac{P_s}{\mu c L_h g (s-r)} = \frac{7209933}{0.8 \times 0.7018 \times 1650.45 \times 9.81 \times \frac{1}{1100}} = 0.72$$

Hilroy

3 Question 3

$$\text{Wetted perimeter} = \pi \times d \times \frac{360 - 40}{360} = 1.396 \text{ m}$$

$$\text{Cross-sectional area} = \frac{1}{4} \pi d^2 \cdot \frac{360 - 40}{360} + \frac{1}{2} r^2 \sin(40^\circ) = 0.1946 \text{ m}^2$$

$$\text{Hydraulic radius (HR)} = \frac{\text{Cross-sectional area}}{\text{Wetted perimeter}} = \frac{0.1946}{1.396} = 0.14 \text{ m}$$